



Department of Mechanical Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Mechanical Engineering

Course Code & Title : CME362 Energy Conservation in Industries

Name of Faculty member : Mr.R.Arun Kumar, AP(SG)/Mech

Innovative Practice Description

- **Unit / Topic** : Unit I / Energy Auditing
- **Course Outcome** : CO6
- **Topic Learning Outcome:** TLO 16
- **Activity Chosen** : Auditing exercise
- **Justification:**
 - A physical energy auditing exercise will give the students a better clarity on the topic and will help them in doing energy auditing in the future if required.
- **Time Allotted for the Activity:** A week / Individually
- **Details of the Implementation:**
 1. Students were asked to inventorize the electrical appliances in their respective homes
 2. They were asked to record the utilization pattern of each electrical appliances
 3. Students have to calculate the electricity consumption of each appliances and the expenditure.
 4. Suggestions for the energy saving opportunities to be listed.
 5. Economic viability to be presented
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PO11	PO12
CO6	3	3	3	2	3	3	3

(2 – Moderate 3 – High)

Note: Justification in Course plan



• Images / Screenshot of the practice:

ASSIGNMENT -1

ENERGY CONSERVATION IN INDUSTRIES

SUBJECT CODE :- CME362

ENERGY AUDITING IN HOME

NAME:- M.SUBBIAH@SELVAM
REG NO:- 95362114051
SUBMISSION DATE:- 31/08/23

ENERGY AUDIT REPORT

Name of the Auditor : M.SUBBIAH@SELVAM
Place of Audit : HOME
Address of Audit :
Date of Audit : 12-08-2023
Type of Audit :

Major Utilities in Home:

Utilities power consumed for an hour for quantity of one.

Sl. No.	Utility	Quantity	Power Consumption Per hour (W-hr)
1	Tube light	5	55
2	Fan	4	75
3	Exhaust Fan	1	40
4	Mixer	1	750
5	Graviser	1	2000
6	Iron Box	1	1000
7	Air conditioner	1	3000
8	Air cooler	1	200
9	Bulb	6	10
10	Washing Machine	1	2000
11	Motor	1	746

Now the power consumed for the running hours for the utilities is being calculated.

Sl. No.	Utility	Quantity	Running Time/Day	Power Consumption per day (W-hr)
1	Tube light	5	5HRS	275
2	Fan	4	15HRS	1125
3	Exhaust Fan	1	2HRS	80
4	Mixer	1	0.5HRS	375
5	Graviser	1	0.5HRS	1000
6	Iron Box	1	0.5HRS	500
7	Air conditioner	1	10HRS	30000
8	Air cooler	1	2HRS	400
9	Bulb	6	2HRS	20
10	Washing Machine	1	2HRS	2000
11	Motor	1	1HRS	746

Scope of Power Consumption:

After implementing the above steps, the power saved by the utilities are to be calculated.

Sl. No.	Utility	Replaced by	Quantity	Running Time/Day	Power Consumption per hour	Power consumption per day	Power Saved per day
1	Tube Light	Led light	3	5HRS	13W	65W	215W
2	Fan	Super fans	4	20HRS	33W	700W	800W
3	Bulb	Led bulb	4	1HRS	8W	24W	6W
4	Kettle						
5	Iron Box						
6	Washing Machine						

Power saved and cost benefits:

BY REPLACING THE NEW TECHNOLOGY WHICH IS MORE EFFECTIVE BY FITTING THOSE ELECTRICALS WE CAN REDUCE THE COST AND POWER CONSUMED.SO THAT WE CAN REDUCE OUR ELECTRICITY BILL .

REFERENCES:https://ijartie.com/AdminUploadPdf/POWER_UTILISATION_CALCULATION_AND_ENERGY_AUDIT_ijartie7709.pdf

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

Based on the feedback collected from the students, they expressed their satisfaction about this activity, motivation towards self-learning energy auditing which may help them in the industries.

❖ Benefit of the practice:

- ✓ Students had hands on experience on energy auditing.
- ✓ It helped them in understanding the steps of energy audit through practice
- ✓ Most of the students reported that, this activity helped them to have a deep understanding the energy conservation opportunities in their respective home

❖ Challenges faced in implementation:

- Hostellers were unable to submit the report as they were unable to monitor their respective homes electricity consumption pattern.